Biochemistry Cume

September 9, 2016

100 points total. Point totals are given next to each question.

1. Examine the amino acid sequence below.

Gly-Glu-Ala

Draw the complete structure, with correct ionization(s) for the tripeptide at pH 2.0, pH 7.0, and pH 10.0. (30 points)

2. Define isoelectric point? How can a protein have an isoelectric point? (20 points)

3. A) How many different tripeptides can be made using only the naturally-occurring amino acids? B) How many tripeptides can be made using G, D, and H? Provide the sequences of 3 such tripeptides, in 3-letter code format, and give their overall charge at pH 7. (40 points)

4. Based on your knowledge of the chemical properties of amino acid side chains, list any amino acid substitution(s) for Val in the 1° structure of a protein that would least likely change the character of the protein. (10 points)